

Architecture Selection and Justification

Architecture Selection: We chose **Layered Architecture** for the Electronic Store Inventory Management System.

Architectural Choice: The system is structured into three layers - a Presentation Layer (User Interface Component), a Business Layer (Inventory Manager, Sales/Billing, and Authentication Components), and a Data Layer (Database Component). Components primarily communicate across defined layer boundaries through clearly defined interfaces, while business components may interact with one another where required by the system workflow.

Reason 1 - Matches System Complexity: Our system has well-defined, sequential business processes (login → search/select product → process sale → update stock → generate invoice) that map naturally onto horizontal layers. Unlike a high-traffic system that needs independent scaling of services (better suited to Microservices), our inventory system is used by a single store with a small number of concurrent users (NFR-006: up to 50 concurrent users), making the operational simplicity of a layered design more appropriate than the added complexity of Microservices.

Reason 2 - Supports Modular Maintainability (NFR-008): Since our requirements explicitly call for modular architecture so that "inventory, billing, and reporting modules can be updated independently" (NFR-008), the layered approach allows each Business Layer component (Inventory Manager, Sales/Billing, Authentication) to be modified or extended without altering the Presentation or Data layers, satisfying this requirement directly.

Security Advantage: By isolating the Authentication Component as a distinct Business Layer service that all Presentation Layer interactions rely on for session validation, the architecture enforces centralized access control at the point of session establishment (login), while individual business components (Inventory Manager, Sales/Billing) independently enforce role checks on each request via NFR-003. This directly supports NFR-003 (RBAC enforcement on all endpoints) and NFR-005 (secure hashing) - the UI layer never accesses the database directly, reducing the attack surface for unauthorized data access.

Performance Benefit: Because all business logic (stock validation, invoice calculation, role checks) is centralized in the Business Layer and only one Data Layer component handles all database I/O, we can optimize database queries and caching in a single place. This directly supports NFR-001 (process a sale transaction within 2 seconds) by avoiding redundant or scattered database calls that a distributed (Microservices) architecture would introduce through network latency.